

SSGMCE	SHRI SANT GAJANAN MAHARAJ COLLEGE OF ENGG., SHEGAON - 444 203.		LABORATORY MANUAL	
	PRACTICAL EXPERIMENT INSTRUCTION SHEET			
	EXPERIMENT TITLE:	TO PERFORM DATA ANALYSIS USING MS-EXCEL.		
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Aim: To perform data analysis using MS-Excel.

Facilities: Computer with MS-Excel installed, sample dataset.

Theory: There are various powerful features that MS-Excel offers to analyze the data. Following are some of these most commonly required features:

A. Sort: We can sort our Excel data on one column or multiple columns, in ascending or descending order.

One Column:

1. Click any cell in the column you want to sort.
2. To sort in ascending order, on the Data tab, in the Sort & Filter group, click AZ.
Note: to sort in descending order, click ZA.

Multiple Columns:

1. On the Data tab, in the Sort & Filter group, click Sort. The Sort dialog box appears.
2. Select "CN1" from the 'Sort by' drop-down list.
3. Click on Add Level.
4. Select "CN2" from the 'Then by' drop-down list.
5. Click OK

Custom Sort:

1. Click any cell inside the data set.
2. On the Data tab, in the Sort & Filter group, click Sort. The Sort dialog box appears.
3. Select "CN1" from the 'Sort by' drop-down list.
4. Select Custom List from the 'Order' drop-down list. The Custom Lists dialog box

appears.

5. Type the list entries.
6. Click OK.
7. Click OK again.

Sort by Color:

1. Click any single cell inside a data set.
2. On the Data tab, in the Sort & Filter group, click Sort. The Sort dialog box appears.
3. Sort by "CN1", sort on Cell Color, and select "Color1" for the first level.
4. Click 'Copy Level' two times and select the other colors.
5. Click OK.

B. Filter: Filter our Excel data if we only want to display records that meet certain criteria.

1. Click any single cell inside a data set.
2. On the Data tab, in the Sort & Filter group, click Filter. Arrows in the column headers appear.
3. Click the arrow next to "CN1".
4. Click on Select All to clear all the check boxes, and click the check box next to "Data1".
5. Click OK.
6. Click the arrow next to "CN2".
7. Click on Select All to clear all the check boxes, and click the check box next to "Data2".
8. Click OK.
9. To remove the filter, on the Data tab, in the Sort & Filter group, click Clear. To remove the filter and the arrows, click Filter.
10. For another way to filter data quickly, select a cell.
11. Right click, and then click Filter, Filter by Selected Cell's Value.
12. Select another cell in another column to further filter this data set.

C. Conditional Formatting: Conditional formatting in Excel enables you to highlight cells with a certain color, depending on the cell's value. We can highlight cells that are greater than a value, less than a value, between two values, equal to a value, cells that contain specific text, dates (today, last week, next month, etc.), duplicates or unique values.

1. Select a range of values.
2. On the Home tab, in the Styles group, click Conditional Formatting.
3. Click Highlight Cells Rules, Greater Than.
4. Enter a value and select a formatting style.

5. Click OK.

Clear Conditional Formatting:

1. Select the formatted range of values.
2. On the Home tab, in the Styles group, click Conditional Formatting.
3. Click Clear Rules, Clear Rules from Selected Cells.

D. Charts:

1. Select a range of values.
2. On the Insert tab, in the Charts group, click the Line symbol.
3. Click Line with Markers.
4. Enter the chart title.

Change Chart Type:

1. Select the chart.
2. On the Design tab, in the Type group, click Change Chart Type.
3. On the left side, click Column, Clustered Column.
4. Click OK.

Switch Row/Column:

1. Select the chart.
2. On the Design tab, in the Data group, click Switch Row/Column.

E. Tables:

Insert Table:

1. Click any single cell inside the data set.
2. On the Insert tab, in the Tables group, click Table.
3. Excel automatically selects the data. Check 'My table has headers' and click on OK.
4. Excel creates a nicely formatted table. This may still seem like a normal data range but many powerful features are now just a click of a button away.

Sort Table:

1. Click the arrow next to "CN1" and click Sort Smallest to Largest.
2. Click the arrow next to "CN2" and click Sort A to Z.

Filter Table:

1. Click the arrow next to "CN1" and only check "Data1".

Total Row:

1. Select a cell inside the table.
2. On the Design tab, in the Table Style Options group, check Total Row.

3. Click any cell in the last row to calculate the Total (Average, Count, Max, Min, Sum etc.) of a column.

F. Analysis ToolPak:

The Analysis ToolPak is an Excel add-in program that provides data analysis tools for financial, statistical and engineering data analysis. There are various data analysis options available with this program, such as, correlation, covariance, descriptive statistics, exponential smoothing, f-test two sample for variances, fourier analysis, histogram, moving average, random number generation, rank and percentile.

To load the Analysis ToolPak add-in, execute the following steps:

1. On the File tab, click Options.
2. Under Add-ins, select Analysis ToolPak and click on the Go button.
3. Check Analysis ToolPak and click on OK.
4. On the Data tab, in the Analysis group, you can now click on Data Analysis.

Correlation:

The correlation coefficient (a value between -1 and +1) tells us how strongly two variables are related to each other. We can use the Analysis Toolpak add-in in Excel to find the correlation coefficient between two variables.

A correlation coefficient of +1 indicates a perfect positive correlation. As variable X increases, variable Y increases. As variable X decreases, variable Y decreases.

A correlation coefficient of -1 indicates a perfect negative correlation. As variable X increases, variable Z decreases. As variable X decreases, variable Z increases.

A correlation coefficient near 0 indicates no correlation.

1. If data analysis is already open, select Correlation and click OK.
2. Select a range of values.
3. Check Labels in first row.
4. Select an empty cell as the Output Range.
5. Click OK.

Conclusion: We have successfully performed various data analysis operations on sample dataset using MS-Excel.

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